

# YUVRAJ BAINS

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## Education

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### Carleton University

Ottawa, Ontario

*Bachelor of Engineering: Software Engineering, Co-op*

2021- 2026

- **Relevant Courses:** Data Structures & Algorithms, Object Oriented Programming, Operating Systems, Web-Based Systems, Computer Security, Software Validation, Database Systems

## Work Experience

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### Web Developer

January 2026 – April

*MarinaLifeGroup*

*Remote*

- Built a full-stack **CRM system** managing **500+ client records** using React and Node.js, reducing manual lead tracking time by **40%** and automating workflows across 3 departments.
- Designed and deployed **20+ RESTful API endpoints** connecting the React frontend to backend services, cutting average API response time from **800ms to 250ms** through query optimization and caching.
- Architected **PostgreSQL** and **MongoDB** schemas supporting **10,000+ client records** with sub-second query performance through strategic indexing and schema normalization.
- Implemented **JWT authentication** with role-based access control across 5 user tiers, passing internal security audits and eliminating all unauthorized access incidents in production.

### Full-Stack Developer

February 2025 – May 2025

*Rüpen Networks*

*Remote*

- Collaborated with a **6-person cross-functional team** to build a fintech web platform for international money transfers, delivering the MVP on time within a 4-month sprint cycle.
- Developed **15+ RESTful API endpoints** using Django and Node.js for authentication and transaction workflows, supporting **1,000+ daily test transactions** with zero downtime.
- Integrated **PostgreSQL** to persist user accounts and transaction data across 8 relational tables, reducing data retrieval latency by **35%** through query indexing and optimization.
- Containerized the development environment using **Docker** and standardized **Git/GitHub** workflows, cutting new developer onboarding time.

## Projects

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### MacroSpots – Macro-Friendly Food Finder | *JavaScript, Leaflet.js*

- A map-first mobile web app that ranks fast food options by a real-time Macro Fit Score, featuring dietary quick filters and color-coded map pins.
- Engineered a **0–100 Macro Fit Score algorithm** weighting 4 macros (calories, protein, carbs, fat) across 3 priority modes (high-protein, low-cal, low-fat), with results rendered as 3-tier color-coded pins on a **Leaflet.js map**.
- Designed a **modular ES6 architecture** with a custom **Pub/Sub state store** (per-key and wildcard listeners) and built a 3-state draggable bottom sheet supporting both touch and mouse input.

### Scrabble Game | *Java, Java Swing*

- A Java Swing implementation of Scrabble featuring a 15x15 drag-and-drop board, AI opponents, dictionary validation, save/load, and undo/redo.
- Built a full **MVC architecture** in **Java Swing** across **18 classes** (Model, View, Controller, AIPlayer, Board, Bag), validating word placements against a **9,999-word dictionary** with live Scrabble-rule scoring.
- Implemented **undo/redo via a dual-stack pattern** and **save/load using Java Serializable**, covered by **12 JUnit tests** across game logic, AI behavior, and board state.

### Online Book Store System | *Java, Spring Boot*

- A full-stack online bookstore built with **Spring Boot** featuring shopping cart, order processing, payment handling, and review management.
- Designed RESTful endpoints using a **layered architecture** (controller/service/repository) across **15+ classes** and integrated **JPA repositories** for persistence.
- Configured **Spring Security** with role-based access control for 3 user tiers (customer/admin/staff), blocking **100%** of unauthorized requests during penetration testing.

## Technical Skills

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**Programming Languages:** C, Java, JavaScript, HTML, CSS, Python, SQL

**Frameworks & Libraries:** Spring Boot, Spring MVC, Hibernate, React, Django, Node.js, Java Swing, Leaflet.js, PostgreSQL, MySQL, MongoDB

**Tools & Technologies:** Linux, Git, GitHub, Docker, JUnit, Maven, Azure